

Fundamental Diagrams

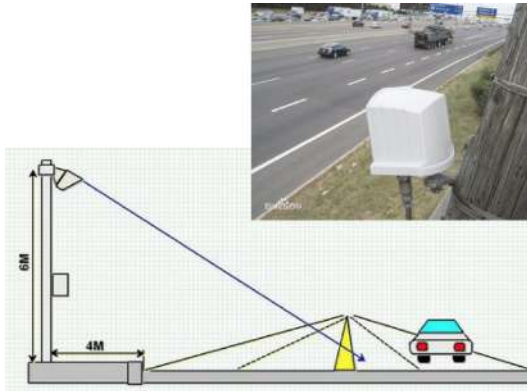
Zhengbing He, Ph.D.

<https://www.GoTrafficGo.com>

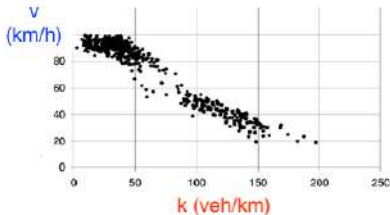
January 2, 2026

Observations

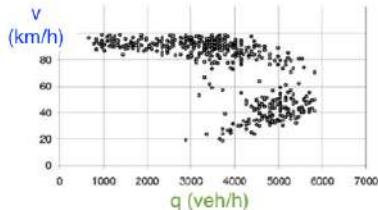
- Let us monitor traffic (e.g.,) at a cross-section
- Compute flow (q), density (k), speed (v), every (e.g.,) 1 min
- Obtain a set of $\langle q, k, v \rangle$ every 1 min
- Last for (e.g.,) a day



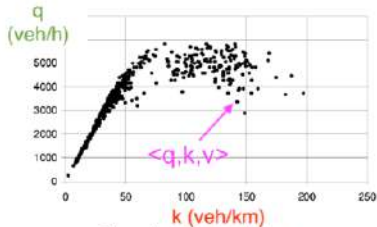
Observations



Observation spots in a v-k diagram



Observation spots in a v-q diagram



Observation spots in a q-k diagram

Cassidy (1998) showed that carefully chosen regions in time-space where traffic conditions are stationary lead to well-defined and scatter-free empirical fundamental diagrams*.

* Cassidy M. Bivariate relations in nearly stationary highway traffic. Transportation Research Part B. 1998.



Pergamon

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BIVARIATE RELATIONS IN NEARLY STATIONARY HIGHWAY TRAFFIC

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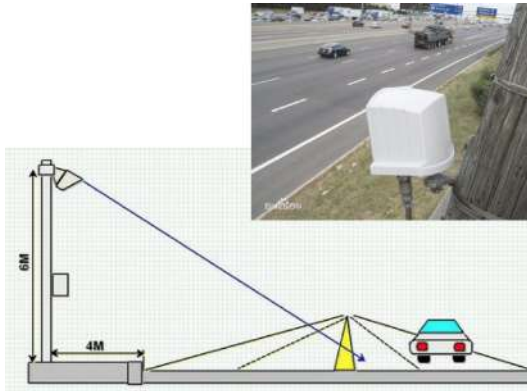
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(Received 18 November 1996; in revised form 20 April 1997)

Abstract—This paper demonstrates that reproducible bivariate relations exist among traffic variables, such as flow and occupancy, when traffic conditions are approximately stationary. The inspection of cumulative curves of vehicle arrival number and vehicle occupancy has revealed that sustained periods of nearly stationary conditions do arise in the traffic stream. By plotting average values of the data corresponding to each nearly stationary condition, well-defined relations are observed. These scatterplots of near-stationary data are contrasted with plots of data that were measured over consecutive time intervals of fixed duration and this reveals that data from certain conditions do not necessarily fall on a curve describing near-stationary traffic.

Observations

- Let us monitor traffic (e.g.,) at a cross-section
- Compute flow (q), density (k), speed (v), every (e.g.,) 1 min
- Obtain a set of $\langle q, k, v \rangle$ every 1 min
- Last for (e.g.,) a day
- Carefully pick stationary traffic out

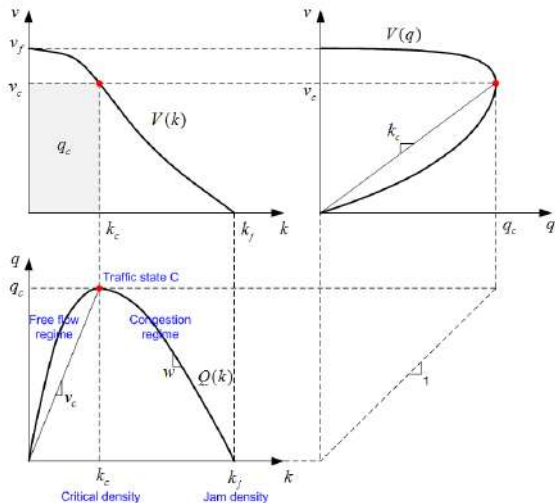


Fundamental Diagrams (FD)

Abstracting from the inhomogeneous and non-stationary traffic, we describe the empirical characteristics using an **equilibrium (stationary) relationship**, which is called the **the fundamental relationship**,

$$q = kv$$

and describe graphically by three diagrams, i.e., **the fundamental diagrams**.



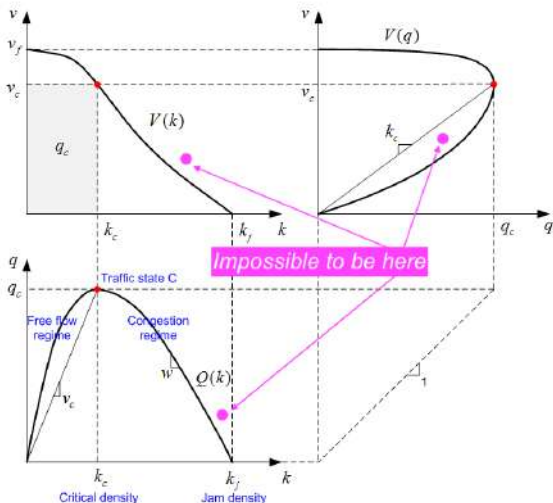
Fundamental Diagrams (FD)

• Traffic states:

Theoretically speaking, traffic is always in a specific **state** that is characterized by (or a result of the balance of)

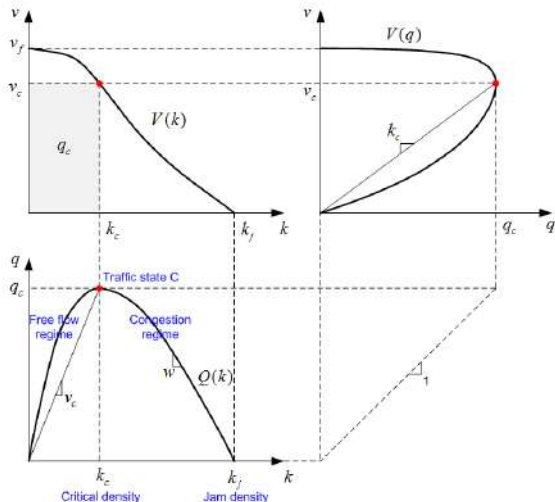
- flow: q
- density: k
- speed: v
space-mean

A traffic state can only appear on the fundamental diagrams



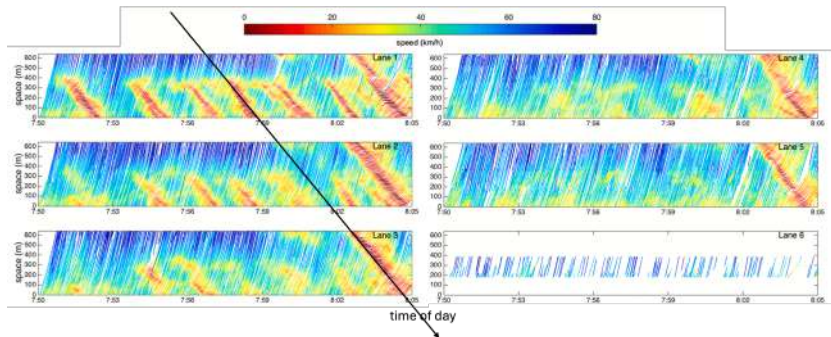
Fundamental Diagrams (FD)

- **Traffic states:**
 q, k, v
- **Free-flow branch/condition**
- **Congestion branch/condition**
- **Critical states:**
 q_c, k_c, v_c
- **Wave speed:** ω



Empirical Trajectories and Waves

Next Generation Simulation (NGSIM) trajectories



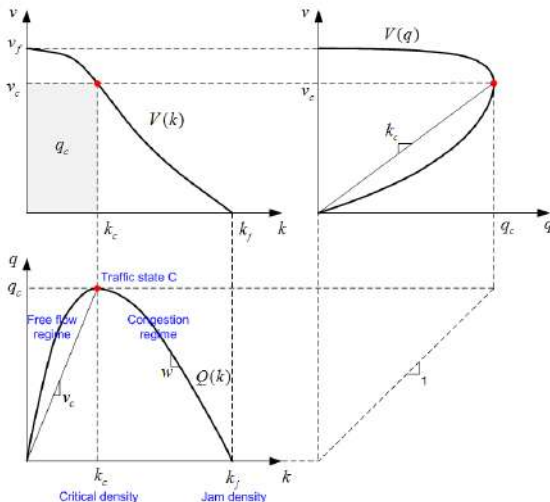
Fundamental Diagrams (FD)

Video: Real-World Traffic and Waves



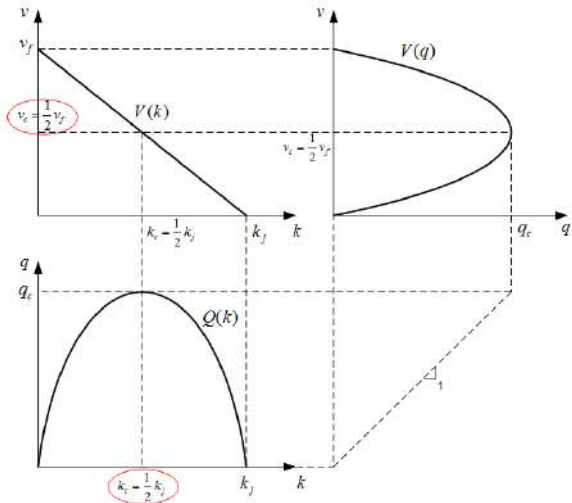
Fundamental Diagrams (FD)

- **Capacity:** $q_c = 2500$ (veh/h)
- **Jam density:** $k_j \approx 150$ (veh/km)
- **Critical density:** $k_c \approx k_j/5$
- **Free flow speed:** $v_f \approx 1.1 v_{lim}$
- **Wave speed:** $\omega \approx 15$ (km/h)



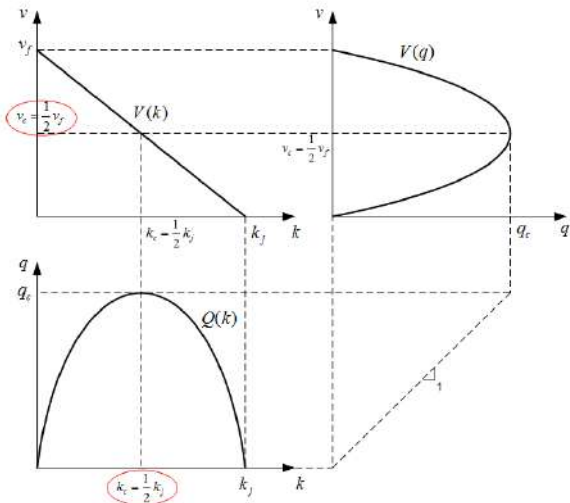
Greenshield Fundamental Diagram

- Proposed by Bruce Greenshield in 1935, his fundamental diagram characterizes a linear relationship between traffic density and speed, from which flow is derived



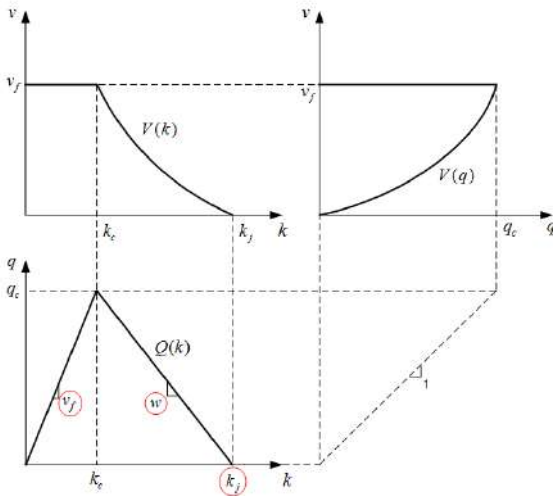
Greenshield Fundamental Diagram

- $v = V(k) = \frac{v_f}{k_j}(k_j - k)$
- $q = V^{-1}(q)$
 $= k_j v (1 - \frac{v}{v_f})$
- $q = Q(k) = \frac{v_f}{k_j} k(k_j - k)$
- **Pros:** One equation



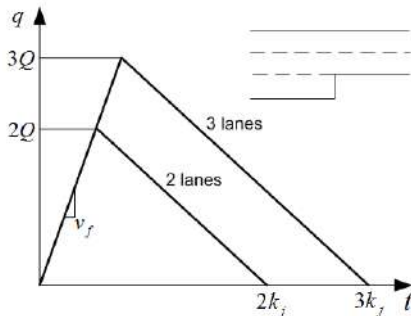
Triangular Fundamental Diagrams (TFD)

- Determining the fundamental relationship only need **three parameters**: v_f , ω , k_j , which is very convenient in traffic modeling.
- $q = \min\{v_f k, -\omega(k - k_j)\}$
- Question:
Critical states: k_c , q_c ?



Practice: Lane Drop

When an accident happens, a lane of a three-lane road is blocked. Try to draw and compare the (T)FDs of the **two-lane** and **three-lane** roadway.

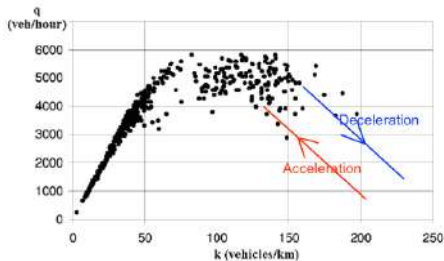


$F(k,t)$: lane drops/additions

Hysteresis

Hysteresis

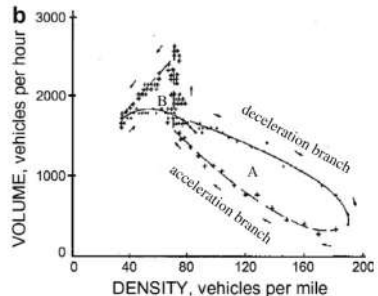
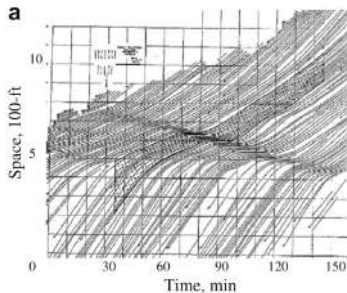
- Dictionary: the **lag** in response exhibited by a body in reacting to changes in the forces.
- The first theory for explaining hysteresis is due to Newell (1962). He conjectured the existence of **two different congested branches** in the fundamental diagram. He argued that in the flow-density diagram, the deceleration branch should be **above** the acceleration branch.



Observation spots in a q-k diagram

Hysteresis

- Zhang (1999) postulated that these branches may **intersect and reverse** the relative position of acceleration and deceleration branches, possibly describing **multiple loops**.



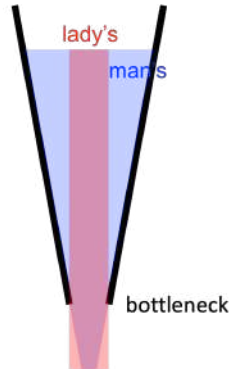
Hysteresis shown in Treiterer and Myers (1974)

Capacity Drop

Capacity Drop

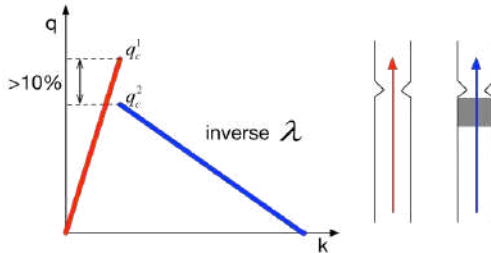
See Video: *Rice Experiment*

Peak shaving and valley filling



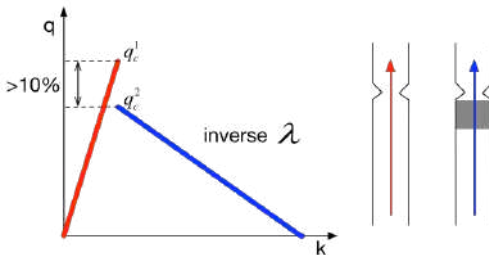
Capacity Drop

- Observation:
 - No congestion (Lady's): **higher** capacity;
 - After being congested (Man's): **lower** capacity;
- Once being congested, the traffic **cannot completely** restore the capacity.



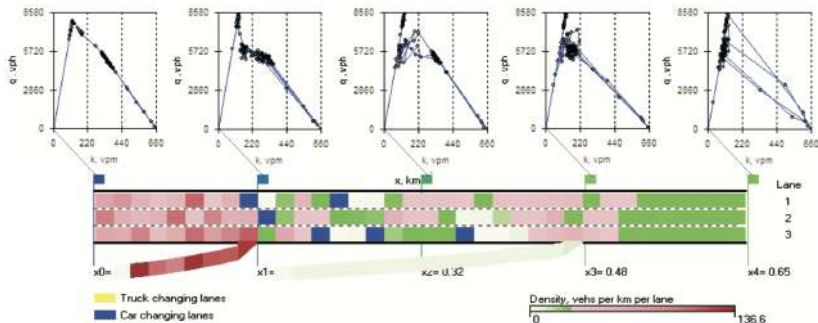
Capacity Drop

- Capacity drop is the **major hazard** of an active bottleneck
- For example, the maximum flow (capacity) of the lane is 2500 veh/h, but only 2000 veh/h can pass through **when the congestion forms** (e.g., lane drop, ramps). The efficiency of traffic is thus **reduced drastically**.



Capacity Drop

Capacity drop arises together with the activation of bottlenecks.



J. A. Laval, "Linking synchronized flow and kinematic waves," in Traffic and Granular Flow'05. 2007

Thank you!